

Affostruction: 3D Affordance Grounding with Generative Reconstruction

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<https://chrockey.github.io/Affostruction/>

Abstract

This paper addresses the problem of affordance grounding from RGBD images of an object, which aims to localize surface regions corresponding to a text query that describes an action on the object. While existing methods predict affordance regions only on visible surfaces, we propose Affostruction, a generative framework that reconstructs complete object geometry from partial RGBD observations and grounds affordances on the full shape including unobserved regions. Our approach introduces sparse voxel fusion of multi-view features for constant-complexity generative reconstruction, a flow-based formulation that captures the inherent ambiguity of affordance distributions, and an active view selection strategy guided by predicted affordances. Affostruction outperforms existing methods by large margins on challenging benchmarks, achieving 19.1 aIoU on affordance grounding and 32.67 IoU for 3D reconstruction.

1. Introduction

Robotic manipulation requires understanding not only what objects a robot observes, but also how it can interact with them. Such functional properties—affordances [8]—are typically predicted from complete 3D shapes [5, 26, 43]. In practice, however, robots observe objects through RGBD cameras from limited viewpoints, resulting in partial observations with significant occlusions. A robot viewing a mug from the front, for instance, needs to reason about the occluded handle for grasping. Existing open-vocabulary affordance grounding methods [17, 28, 31, 38] operate directly on partial point clouds, predicting affordances only on visible surfaces. Meanwhile, multi-view reconstruction methods [11, 24, 39, 41, 42, 48] and 3D generation methods [12, 20, 45] address geometry but not functional affordances. Reconstruction methods recover only observed surfaces, while generation methods can complete unseen regions but operate on single RGB images without leveraging the depth available in robotic settings. Grounding affordances on occluded surfaces thus requires both completing

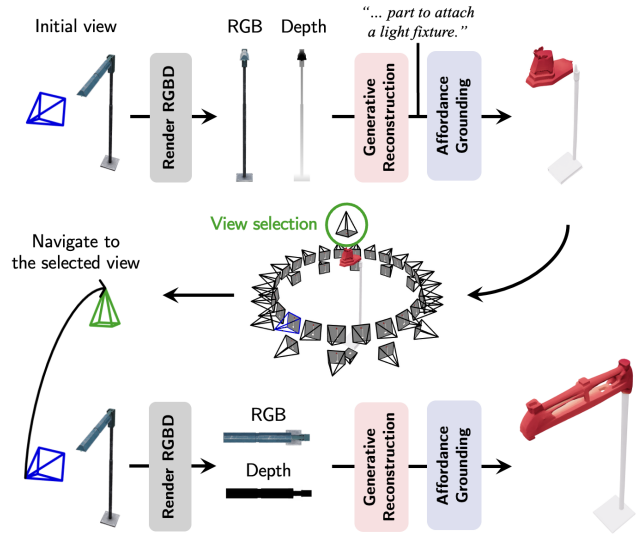


Figure 1. **Affostruction.** Given an initial RGBD observation (blue camera) where functional regions for an affordance query (e.g., “attach a light fixture”) are only partially visible or heavily occluded, we reconstruct the complete 3D geometry in a generative manner – estimating unobserved surfaces – and ground an affordance region on the full shape effectively. Building on this, an affordance-driven active view selection strategy identifies the most informative next viewpoint (green camera). The additional observation acquired from this selected view further refines both the 3D reconstruction and the affordance grounding of the target region.

geometry from depth-conditioned observations and predicting functional properties on the reconstructed shape.

We introduce **Affostruction**, a generative framework that reconstructs complete object geometry from partial RGBD observations and grounds affordances on the full shape including unobserved regions. Two properties of the robotic setting further shape our design. First, robotic scenarios naturally provide multi-view RGBD sequences as cameras move or objects are manipulated. We exploit this through sparse voxel fusion of DINOv2 [30] features that aggregates information from multiple views while maintaining constant computational complexity, enabling gen-

erative reconstruction that extrapolates unseen geometry. Second, affordances are inherently ambiguous—multiple valid regions exist for the same query (*e.g.*, “grasp”). We address this through flow-based generation that learns to model distributions over affordance heatmaps rather than relying on discriminative predictions, naturally capturing the multi-modal nature of functional interactions. Moreover, the predicted affordances on reconstructed geometry can guide active viewpoint selection, prioritizing views that observe functional regions to progressively improve target region reconstruction and grounding.

We build upon TRELLIS [45], a foundation model for 3D generation trained on large-scale 3D datasets. TRELLIS provides strong priors for completing 3D structures, but processes only single RGB images without depth or camera extrinsics, and generates visual appearance rather than functional affordances. Without depth, it cannot resolve structural details where textures are similar but geometry differs; without camera extrinsics, reconstructions are canonicalized and may not align with the actual input orientation. We extend it with two components: multi-view sparse voxel fusion that aggregates depth-conditioned DINOv2 [30] features from multiple RGBD views with known camera poses into a constant-size representation, and a flow-based affordance module that generates heatmaps conditioned on text queries over the reconstructed sparse voxels. These extensions enable Affostruction to ground affordances on complete geometry from multi-view RGBD observations while leveraging the foundation model’s geometric priors.

Experiments on the Affogato [15] benchmark show that Affostruction achieves 32.67 IoU for 3D reconstruction (54.8% improvement over the state of the art) and 19.1 aIoU for affordance grounding on complete geometry (40.4% improvement). When operating from partial RGBD observations, our method reconstructs complete geometry and grounds affordances on the full shape, including occluded regions, from as few as a single view. We further leverage predicted affordances to guide active viewpoint selection, prioritizing views that maximize visibility of functional regions. With just one additional view, this strategy achieves $2.0\times$ faster improvement over sequential baselines, providing clear gains under limited view budgets.

Our contributions are as follows:

- **Generative multi-view reconstruction:** We propose sparse voxel fusion of depth-conditioned DINOv2 features with camera extrinsics that enables constant-complexity generative reconstruction aligned with the actual object orientation, extrapolating complete geometry from partial RGBD observations.
- **Flow-based affordance grounding:** We introduce a flow-based generative model that predicts affordance heatmaps on reconstructed geometry conditioned on natural language queries, capturing the inherent multi-

modality of functional interactions.

- **Affordance-driven active view selection:** We select next-best views based on affordance predictions to improve functional region coverage, achieving efficient reconstruction and grounding under limited view budgets.

2. Related work

3D reconstruction. Multi-view 3D reconstruction recovers geometry from multiple RGB or RGBD images. Classical TSDF fusion methods [3, 14, 27, 29] accumulate depth measurements into volumetric representations, while learning-based approaches such as MVSNet [48] and its variants [6, 9, 49] predict depth through cost volume matching. Neural implicit methods [39, 41] learn continuous surface representations from multi-view images but require per-scene optimization, while feed-forward models such as DUST3R [42] and MAST3R [16] predict dense correspondences without camera calibration, enabling direct generalization to unseen scenes. Despite these advances, all reconstruction methods are inherently limited to observed surfaces and do not extrapolate geometry to unseen regions. Shape completion methods such as OctMAE [11] and MCC [44] extend reconstruction to unobserved regions through learned priors, but none of these methods predict affordances on the recovered surfaces.

3D generation. Image-conditioned 3D generation methods [12, 19, 20, 40, 46] produce complete shapes from single images via feed-forward prediction or multi-view diffusion. Sparse voxel generation methods such as TRELLIS [45], XCube [33], and 3DTopia-XL [1] decompose the problem into structure and appearance stages in 3D latent space. Multi-view diffusion methods [22, 35, 47] concatenate per-view tokens, resulting in $O(N)$ token growth that limits inputs to a few views at reduced resolution. These methods rely solely on RGB inputs without using depth to resolve geometric details where textures are ambiguous, nor do they use camera extrinsics, producing canonicalized outputs that may not align with the actual scene. None of these methods accept multi-view RGBD inputs or predict functional properties on the generated shapes.

3D affordance grounding. 3D AffordanceNet [5] introduced the first 3D affordance grounding benchmark with fixed labels, and interaction-based methods [26, 43] learn affordances through simulated contacts on complete shapes. Open-vocabulary methods [17, 28, 31, 51] integrate point cloud features with vision-language embeddings to handle broader queries, and OVA-Fields [38] extends this to 3D fields. Affogato [15] scales annotation through automated data generation, and its grounding module Espresso-3D handles diverse open-vocabulary queries. All these methods assume complete point clouds or predict affordances only on observed surfaces, and their discriminative formu-

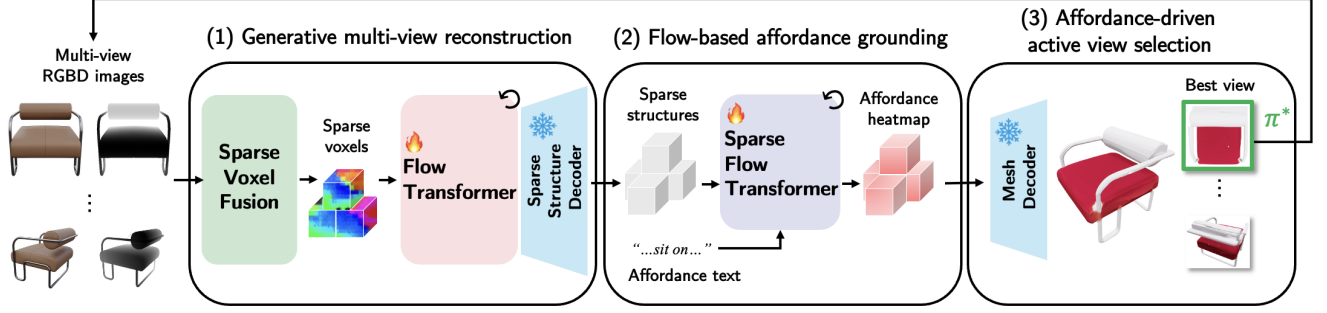


Figure 2. **Affostruction overview.** Our approach consists of three stages. (1) Generative multi-view reconstruction: DINOv2 [30] features from RGBD views are fused into sparse voxels using depth and camera parameters, and a flow transformer extrapolates complete structure from partial observations, decoded via a pretrained decoder [45]. (2) Flow-based affordance grounding: a sparse flow transformer conditioned on CLIP [32]-encoded text generates affordance heatmaps over reconstructed geometry. (3) Affordance-driven active view selection: next-best viewpoints maximize visibility of high-affordance regions, and a mesh decoder [45] produces the final 3D mesh.

lations produce single-mode predictions that cannot capture the multi-modal nature of affordances. Our method addresses both gaps: it completes geometry from multi-view RGBD observations via sparse voxel fusion with $O(1)$ token complexity, and grounds affordances through flow-based generation that models multi-modal distributions.

3. Affostruction

Given multi-view RGBD images with camera parameters, Affostruction reconstructs complete 3D object geometry—including unobserved regions—and grounds open-vocabulary affordances on it.

As illustrated in Fig. 2, our approach consists of three stages: (1) Generative multi-view reconstruction (Sec. 3.2) fuses features from multiple RGBD views into sparse voxels using depth and camera parameters, extrapolating complete 3D structure from partial observations. (2) Flow-based affordance grounding (Sec. 3.3) predicts affordance distributions over the reconstructed geometry conditioned on natural language queries. (3) Affordance-driven active view selection (Sec. 3.4) leverages affordance heatmaps to select viewpoints that prioritize functional regions.

3.1. Preliminaries: TRELLIS

We build upon TRELLIS [45], a two-stage rectified flow framework [21] for 3D generation in sparse latent space:

Stage 1: flow transformer denoises a dense latent tensor $\mathbf{X} \in \mathbb{R}^{r^3 \times C}$ ($r=16$) conditioned on DINOv2 [30] patch features $\mathbf{C}$ extracted from a single RGB image. A pretrained sparse structure VAE decoder [45] then converts the denoised tensor into sparse structure $\{\mathbf{p}_i\}_{i=1}^L$, a set of occupied voxel positions where $L \ll r^3$.

Stage 2: sparse flow transformer denoises a sparse latent tensor initialized at the occupied positions from Stage 1, conditioned on image features, producing structured latent

features $\{\mathbf{z}_i\}_{i=1}^L$. A pretrained 3D decoder [45] then renders these into the final 3D representation (e.g., mesh, 3D Gaussians, or radiance fields).

Both stages are trained using the conditional flow matching (CFM) objective [18]:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{X}_0, \epsilon} [\|v_\theta(\mathbf{X}_t, \mathbf{C}, t) - (\epsilon - \mathbf{X}_0)\|_2^2], \quad (1)$$

where $\mathbf{X}_t = (1-t)\mathbf{X}_0 + t\epsilon$ interpolates between clean data $\mathbf{X}_0$ and noise $\epsilon \sim \mathcal{N}(0, I)$ with timestep $t \in [0, 1]$, and v_θ predicts the velocity field.

Limitations. While effective for single-image 3D generation, TRELLIS has two limitations: (1) it accepts only a single RGB image without depth, limiting accurate structure reconstruction from partial observations; (2) the sparse flow transformer produces visual appearance, not functional affordances. We address (1) by extending the flow transformer to fuse multi-view RGBD inputs (Sec. 3.2) and (2) by training a new sparse flow transformer to generate affordance heatmaps from text queries (Sec. 3.3).

3.2. Generative multi-view reconstruction

We extend TRELLIS’s flow transformer to support multi-view RGBD inputs through *multi-view sparse voxel fusion*. Rather than processing views independently, we aggregate features in 3D using depth guidance before the transformer. This enables generative reconstruction that extrapolates complete geometry from partial observations while maintaining constant token complexity.

Sparse voxel fusion conditioning. We represent each view as $\mathcal{V}_i = (I_i, D_i, K_i, T_i)$, where I_i is an RGB image, D_i is a depth image, K_i is camera intrinsics, and T_i is camera extrinsics. For each view, we extract DINOv2 [30] features from I_i and project them to 3D space: each pixel (u, v) with depth d maps to world coordinates $\mathbf{p} = T_i \cdot K_i^{-1}[u, v, d]^\top$, with its DINOv2 feature $\mathbf{f} = \text{DINOv2}(I_i)_{u,v}$. This produces a sparse voxel representation $\mathbf{V}_i = \{(\mathbf{p}_j, \mathbf{f}_j)\}_{j=1}^{M_i}$ for

view i , where M_i is the number of observed voxels.

To fuse multiple views $\{\mathcal{V}_i\}_{i=1}^N$, we merge their sparse voxel representations. For overlapping voxels at the same position, we average their features; for non-overlapping voxels, we include them in the union. The fused representation $\bar{\mathbf{V}} = \{(\mathbf{p}_m, \bar{\mathbf{f}}_m)\}_{m=1}^M$ is then combined with 3D positional encodings to form the final conditioning signal:

$$\mathbf{C}_{\text{voxel}} = \{\bar{\mathbf{f}}_m + \text{PE}_{3\text{D}}(\mathbf{p}_m)\}_{m=1}^M, \quad (2)$$

where $\text{PE}_{3\text{D}}$ is a standard sinusoidal positional encoding extended to 3D voxel coordinates, concatenating separate sinusoidal encodings for each spatial dimension.

Stochastic multi-view training. Although TRELLIS’s cross-attention conditioning mechanism can handle varying numbers of input tokens at inference, we observe that models trained only on single views exhibit performance degradation when given multiple views (Fig. 4, left). To address this, we employ stochastic multi-view training: randomly sampling 1-8 views with randomized positions during each training iteration. This enables the model to leverage additional observations, showing consistent improvements as more views are added (Fig. 4, right).

Training objective. We feed the conditioning signal $\mathbf{C}_{\text{voxel}}$ to the flow transformer [45] via cross-attention and train with the same rectified flow objective (Eq. (1)), enabling the model to extrapolate complete 3D structure from partial observations at inference.

3.3. Flow-based affordance grounding

Given the reconstructed 3D geometry and a natural language query (*e.g.*, “where to grasp”), we predict affordance heatmaps over the complete shape. Since affordances are multi-modal—a single query admits multiple valid interaction regions—we train a sparse flow transformer from scratch to generate affordance heatmaps as distributions rather than discriminative predictions.

Text conditioning. We encode the text query through a pre-trained CLIP [32] text encoder to obtain text embeddings:

$$\mathbf{C}_{\text{text}} = \text{CLIP}_{\text{text}}(q), \quad (3)$$

where q is the natural language affordance query (*e.g.*, “where to grasp”) and $\mathbf{C}_{\text{text}} \in \mathbb{R}^d$ serves as the conditioning signal. The affordance flow model denoises a sparse noise tensor $\mathbf{A}_1$ initialized at the voxel positions predicted by Stage 1, conditioned on the text embedding $\mathbf{C}_{\text{text}}$.

Training objective. Following the rectified flow formulation, the model predicts velocity fields v_θ that denoise from noise $\epsilon \sim \mathcal{N}(0, I)$ to clean affordance logits $\mathbf{A}_0 \in \mathbb{R}^L$ at L occupied voxels, where the noisy state $\mathbf{A}_t = (1-t)\mathbf{A}_0 + t\epsilon$ with $t \in [0, 1]$. Unlike structure generation which uses MSE loss (Eq. (1)), affordance grounding requires capturing binary occupancy patterns of functional regions. We there-

fore define a binary mask loss that combines binary cross-entropy and Dice loss [25]:

$$\mathcal{L}_{\text{mask}}(\mathbf{A}', \mathbf{A}) = \mathcal{L}_{\text{BCE}}(\mathbf{A}', \mathbf{A}) + \mathcal{L}_{\text{Dice}}(\mathbf{A}', \mathbf{A}), \quad (4)$$

where $\mathbf{A}', \mathbf{A} \in \mathbb{R}^L$ are predicted and ground truth affordance logits, respectively (sigmoid transformations are omitted for brevity but applied within BCE and Dice). This provides both point-wise supervision (BCE) and region-level optimization (Dice). The flow matching objective with binary mask loss is:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{A}_0, \epsilon} [\mathcal{L}_{\text{mask}}(\epsilon - v_\theta(\mathbf{A}_t, \mathbf{C}_{\text{text}}, t), \mathbf{A}_0)]. \quad (5)$$

Only the affordance flow model is optimized while the structure generation model and text encoder remain frozen. At inference, we denoise sampled noise conditioned on text query to generate affordance logits, then apply sigmoid to obtain probability heatmaps.

3.4. Affordance-driven active view selection

Since Affostruction predicts affordances on reconstructed geometry, the predicted heatmaps can guide subsequent view selection. Given current observations, we select the next-best viewpoint that maximizes visibility of high-affordance regions, prioritizing functional areas during multi-view capture.

Candidate view generation. We generate a set of $K = 40$ candidate camera poses $\Pi = \{\pi_1, \dots, \pi_K\}$ uniformly distributed on a hemisphere around the target object.

Affordance visibility metric. Given the sparse structure with affordance heatmap $\{(\mathbf{p}_i, a_i)\}_{i=1}^L$ from Sec. 3.3, we decode it into an affordance-colored mesh $\mathcal{M}$ using the pre-trained TRELLIS [45] mesh decoder. For each candidate pose $\pi_i \in \Pi$, we render this affordance-colored mesh $\mathcal{M}$ to obtain a 2D image A_{render} . The affordance visibility score is simply the sum of heatmap values in the rendered image:

$$\mathcal{S}(\pi_i, \mathcal{M}) = \sum_{u,v} A_{\text{render}}(u, v), \quad (6)$$

where $A_{\text{render}}(u, v)$ is the affordance heatmap value at pixel (u, v) in the rendered image from pose π_i . This metric prioritizes viewpoints that observe high-affordance regions.

Iterative view selection. We select the next viewpoint by maximizing the visibility score:

$$\pi^* = \arg \max_{\pi_i \in \Pi} \mathcal{S}(\pi_i, \mathcal{M}). \quad (7)$$

Starting from an initial view $\mathcal{V}_1$, we select the next-best pose π^* , capture RGBD observation $\mathcal{V}^*$ from π^* , and add it to the view set. This process repeats iteratively, with each round of reconstruction and affordance prediction informing the next viewpoint selection.

Table 1. **3D reconstruction results on Toky4K [37]**. We compare RGB-to-3D generation models and MCC [44], an RGBD-to-3D reconstruction model. Since MCC does not produce mesh outputs, rendering-based metrics (PSNR, LPIPS) are not available.

Method	Depth	Geometry					Appearance	
		IoU $\uparrow$	CD $\downarrow$	F-score $\uparrow$	PSNR-N $\uparrow$	LPIPS-N $\downarrow$	PSNR $\uparrow$	LPIPS $\downarrow$
Shap-E [12]		6.39	0.6724	0.0096	16.16	0.3014	13.07	0.4358
InstantMesh [46]		13.68	0.4063	0.0391	20.46	0.2463	16.59	0.2989
LGM [40]		9.39	0.5660	0.0267	17.03	0.3974	13.59	0.4126
TRELLIS [45]		19.49	0.3694	0.0496	20.96	0.2089	17.61	0.2435
MCC [44]	✓	21.11	0.3299	0.0648	N/A	N/A	N/A	N/A
Affostruction (ours)	✓	32.67	0.2427	0.0997	22.64	0.1421	18.84	0.1922

Table 2. **Complete 3D affordance grounding results on Affogato [15]**. All methods receive ground-truth complete geometry as input. aIoU is the primary metric for spatial localization.

Method	aIoU $\uparrow$	AUC $\uparrow$	SIM $\uparrow$	MAE $\downarrow$
OpenAD [28]	3.1	64.8	0.329	0.150
PointRefer [17]	10.5	76.1	0.405	0.120
Espresso-3D [15]	13.6	79.0	0.429	0.111
Affostruction (ours)	19.1	72.0	0.426	0.217

4. Experiments

We evaluate Affostruction on 3D reconstruction (Sec. 4.2), affordance grounding on complete geometry (Sec. 4.3), and affordance grounding from partial observations (Sec. 4.4). We further analyze multi-view training, active view selection, multi-object scenes, and runtime (Sec. 4.5).

4.1. Setup

Training datasets. For the reconstruction flow transformer, we follow TRELLIS [45] and train on 3D-FUTURE [7], HSSD [13], ABO [2], and Affogato’s train split [15]. We use Affogato’s train split instead of Objaverse-XL [4], which overlaps with Affogato source data, to prevent test set leakage. The affordance flow transformer is trained on Affogato’s train split, which contains affordance annotations.

Evaluation datasets. Following TRELLIS [45], we evaluate 3D reconstruction on 1,250 samples from Toky4K [37], a dataset of 4,000 toy objects with ground truth geometry (sampled object IDs provided in supplementary material). For affordance grounding (both complete and partial settings), we use Affogato’s test split [15]. Partial evaluation uses only the first RGBD view as input.

Implementation details. We train for 1M steps on 8 A100 GPUs with batch size 8 per GPU, using AdamW [23] ($\text{lr} = 10^{-4}$). Visual features are extracted using DINOv2-large [30], and text queries are encoded with CLIP ViT-L/14 [32]. We use classifier-free guidance [10] with 10% unconditional dropout during training. Stochastic multi-view training samples 1-8 views per iteration.

Table 3. **Partial 3D affordance grounding results on Affogato [15]**. Methods without reconstruction predict affordances only on observed surfaces. Two-stage pipelines pair a reconstruction model with a pretrained affordance model.

Method	Recon.	aIoU $\uparrow$	aCD $\downarrow$
OpenAD [28]		0.38	0.4165
PointRefer [17]		0.53	0.3072
Espresso-3D [15]		0.60	0.2885
TRELLIS [45] + OpenAD [28]	✓	1.49	0.1671
TRELLIS [45] + PointRefer [17]	✓	2.05	0.1576
TRELLIS [45] + Espresso-3D [15]	✓	2.23	0.1568
MCC [44] + OpenAD [28]	✓	3.34	0.1503
MCC [44] + PointRefer [17]	✓	4.19	0.1397
MCC [44] + Espresso-3D [15]	✓	4.74	0.1354
Affostruction (ours)	✓	9.26	0.1044

4.2. 3D reconstruction

Metrics. We measure geometric accuracy via volumetric IoU, Chamfer Distance (CD), and F-score ($\tau = 0.05$). For appearance, we render RGB and normal images from a random view ($r = 2$, $\text{FoV} = 40^\circ$) and compute PSNR and LPIPS (-N suffix for normals).

Results. Table 1 compares reconstruction quality on Toky4K. Affostruction achieves 32.67 IoU, outperforming both TRELLIS [45] (19.49, +67.6%) and MCC [44] (21.11, +54.8%). TRELLIS produces high-fidelity generations but, without depth conditioning, suffers from inconsistencies in object orientation and scale relative to ground truth, resulting in poor geometric metrics. MCC addresses this through depth-based reconstruction, achieving better geometric accuracy than TRELLIS across IoU, CD, and F-score, though it does not produce mesh outputs for rendering-based evaluation. Our method surpasses both by combining depth conditioning with flow matching-based generation, showing that generative reconstruction with sparse voxel fusion provides stronger geometric priors than RGB-only generation or discriminative depth-based methods.



Figure 3. **Qualitative results on partial 3D affordance grounding.** Affostruction reconstructs complete geometry and grounds affordances throughout entire objects from single RGBD views. Despite limited observations, our method predicts affordances on occluded regions, demonstrating the ability to reason about 3D functional interactions even when large portions of objects are unobserved.

4.3. Complete 3D affordance grounding

Metrics. Following Espresso-3D [15], we report average Intersection over Union (aIoU), Area Under Curve (AUC), Similarity (SIM), and Mean Absolute Error (MAE). Among these, we consider aIoU as the primary metric: since affordance regions serve as contact points for robotic manipulation, precise spatial localization matters most. Accordingly, we train with standard BCE, whereas prior methods [15, 17] adopt focal BCE, which lowers MAE but at the cost of aIoU.

Results. Table 2 evaluates affordance grounding given complete geometry. We compare against OpenAD [28] and PointRefer [17] using their official implementations, and Espresso-3D [15] reimplemented following the paper. Unlike existing methods [15, 17, 28] that fine-tune vision and text encoders with discriminative objectives, our generative approach produces affordance heatmaps without additional encoder training. While this yields slightly lower AUC (72.0 vs 79.0) and SIM, our method achieves 19.1

aIoU—40.4% better than Espresso-3D (13.6), indicating more precise localization of affordance regions. This indicates that generative modeling of affordance distributions achieves stronger spatial grounding than discriminative approaches, even without task-specific encoder fine-tuning.

4.4. Partial 3D affordance grounding

Metrics. In the partial setting, affordances must be evaluated across the entire 3D geometry including unobserved regions—a requirement for robotic manipulation that needs functional understanding beyond visible surfaces. While prior work [15, 17, 28] evaluates affordances only within observed regions, we introduce metrics that assess affordance prediction on complete 3D geometry. We threshold both predicted and ground truth affordance point clouds $\{(\mathbf{p}'_i, a'_i)\}$ and $\{(\mathbf{p}_j, a_j)\}$ at five probability levels ($\tau \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$) to obtain binary affordance regions, and compute volumetric IoU and Chamfer Distance (CD)

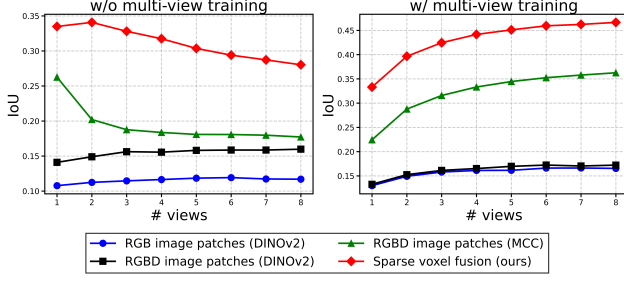


Figure 4. **Impact of multi-view training.** Reconstruction IoU as a function of input views. Single-view trained models (left) show minimal gains from additional views at inference, while stochastic multi-view training (right) enables consistent improvement. Our sparse voxel fusion achieves the best performance in both settings.

between them. We report their averages:

$$\begin{aligned} \text{aIoU} &= \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \text{IoU}(\mathcal{P}'_{\tau}, \mathcal{P}_{\tau}), \\ \text{aCD} &= \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \text{CD}(\mathcal{P}'_{\tau}, \mathcal{P}_{\tau}), \end{aligned} \quad (8)$$

where $\mathcal{P}_{\tau} = \{\mathbf{p}_i \mid a_i \geq \tau\}$ are thresholded point sets and $\mathcal{T} = \{0.1, 0.2, 0.3, 0.4, 0.5\}$. Averaging across multiple thresholds avoids sensitivity to a single binarization criterion. These metrics jointly evaluate two capabilities: accurate 3D reconstruction and precise affordance localization on the reconstructed geometry—high performance requires both, as accurate reconstruction alone is insufficient if affordance predictions are poorly localized, and affordance predictions cannot compensate for inaccurate geometry.

Results. Table 3 evaluates affordance prediction on complete objects from single RGBD views. Existing methods [15, 17, 28] predict affordances only on observed surfaces without reconstruction, yielding 0.38–0.60 aIoU on complete geometry since they cannot account for unobserved regions. Two-stage pipelines using TRELLIS [45] (RGB) or MCC [44] (RGBD) for reconstruction followed by pretrained affordance models substantially improve over this, with the best reaching 4.74 aIoU (MCC+Espresso-3D), indicating that geometric completion is critical for affordance prediction beyond visible surfaces. Affostruction achieves 9.26 aIoU—nearly doubling MCC+Espresso-3D—by performing both tasks in a sparse voxel space. Figure 3 shows qualitative reconstructed affordances.

4.5. Analysis

Stochastic multi-view training. We compare four conditioning approaches using the same flow transformer: (1) *RGB image patches*: DINOv2 features from RGB only, as in TRELLIS [45], (2) *RGBD image patches (DINOv2)*: depth maps processed through DINOv2, (3) *RGBD image*

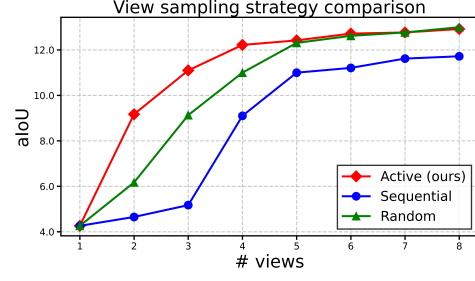


Figure 5. **Quantitative results of active view selection.** Affordance grounding quality (aIoU) as views are incrementally added from minimal affordance visibility. Affordance-driven selection (red) achieves the fastest improvement, sequential sampling (blue) improves slowest due to its fixed trajectory, and random sampling (green) converges with active selection given more views.

patches (MCC): features from pretrained MCC [44] encoder, and (4) *Sparse voxel fusion (ours)*: multi-view DINOv2 features fused in 3D via depth-guided projection. Each is trained with either single-view or stochastic multi-view (1–8 views per iteration) supervision. Figure 4 shows that multi-view training is critical for leveraging additional observations: models trained on single views show minimal improvement or degradation when given multiple inputs at inference, while stochastic multi-view training yields consistent improvement as views increase, plateauing around 6–8 views. Affostruction achieves the best results, with IoU improving from 33.32 (single view) to 46.65 (8 views).

Affordance-driven active view selection. We evaluate whether affordance predictions can guide view selection to prioritize functional regions on the Furnitures subset of Affogato’s test split, where affordance regions are relatively well-defined. To simulate a challenging starting condition, we select the initial view with the lowest affordance visibility using ground truth heatmaps. Given this initial view, we compare three strategies: sequential (predetermined circular trajectory), random (uniform selection), and affordance-driven (ours, selecting views that maximize visibility of predicted high-affordance regions). Figure 5 shows that all methods start from the viewpoint with minimal affordance visibility (~ 4.3 aIoU), and affordance-driven selection achieves the fastest improvement, reaching 9.2 aIoU with one additional view— $2.0\times$ over sequential (4.7) and $1.5\times$ over random (6.2). The advantage persists through 4 views (12.4 vs. 9.1 sequential and 11.0 random). By 8 views, active and random sampling converge (13.3 vs. 13.2 aIoU), while sequential remains behind at 11.7 aIoU, as its predetermined trajectory may miss functional regions. Figure 1 illustrates this process on a fireplace queried for placing firewood logs. From an initial side view where the target region is not visible, the generative prior still produces a reasonable reconstruction and affordance grounding. This

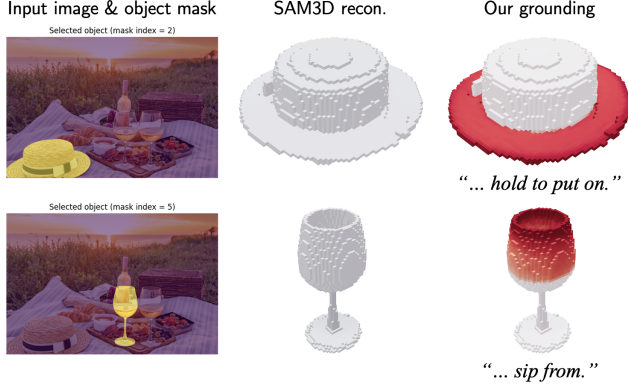


Figure 6. **Extension to multi-object scenes.** Given a multi-object scene, SAM3D [34] reconstructs and segments individual objects, and our method grounds affordances on each object independently.

Table 4. **Runtime and memory comparison.** Average runtime (sec) and peak memory (GB) on Affogato [15], measured on a single RTX A6000 GPU.

Method	Runtime (sec) ↓			Memory (GB) ↓		
	Rec.	Aff.	Total	Rec.	Aff.	Peak
TRELLIS + Esp.-3D	7.91	0.39	8.30	16.90	4.36	16.90
MCC + Esp.-3D	34.98	0.39	35.37	1.39	4.36	4.36
Affostruction (ours)	4.74	2.46	7.20	16.35	1.26	16.35

estimate guides selection of a next best view that better observes the affordance region, and incorporating it improves reconstruction and grounding.

Extension to multi-object scenes. While our work focuses on object-centric settings, we demonstrate extensibility to multi-object scenes by integrating with SAM3D [34]. Given a multi-object scene, SAM3D reconstructs and segments individual objects, and our method then grounds affordances on each object independently. Figure 6 shows that Affostruction successfully grounds affordances on each object in this setting. Since SAM3D does not support multi-view conditioning, integrating our sparse voxel fusion-based conditioning and active view selection with multi-object reconstruction is an interesting future direction.

Runtime and memory. Table 4 compares runtime and peak memory on a single RTX A6000 GPU. Our method achieves the fastest runtime (7.20 sec): sparse voxel fusion conditioning reduces sampling to 5 steps (vs. 25 for TRELLIS), and joint inference avoids the overhead of separate models. MCC processes 64^3 grid points in 2000-query chunks sequentially, resulting in the slowest runtime (35.37 sec) but the lowest peak memory (4.36 GB). Overall, Affostruction achieves the best accuracy (Table 3) with the fastest runtime and comparable memory footprint.

Failure cases. Figure 7 illustrates two failure modes. First, severe occlusion can cause reconstruction errors that prop-

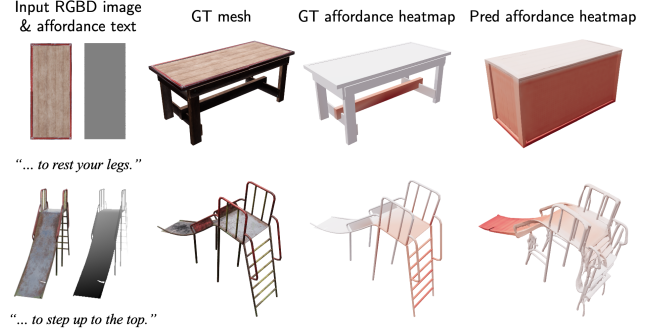


Figure 7. **Failure cases.** (top) Challenging views with severe occlusion lead to reconstruction errors that propagate to affordance predictions. (bottom) Incorrect initial affordance grounding misleads active view selection away from the target region.

agate to affordance predictions (top); conditioning reconstruction on the affordance text query could provide semantic cues that help recover accurate geometry in such cases. Second, incorrect initial affordance grounding can mislead active view selection away from the target region (bottom), as subsequent views are chosen based on erroneous estimates; incorporating error detection strategies from semi-supervised learning [36, 50] to identify and correct unreliable predictions is a promising future direction.

5. Conclusion

We have presented Affostruction, a generative framework that completes object geometry and grounds affordances beyond observed surfaces from partial RGBD observations. Sparse voxel fusion enables constant-complexity multi-view reconstruction, flow matching captures diverse affordance distributions on the reconstructed geometry, and affordance-driven view selection prioritizes functional regions under limited view budgets. These components form a closed loop: affordance predictions guide view selection, which in turn improves both reconstruction and grounding. Experiments demonstrate consistent improvements over existing methods in reconstruction, affordance grounding, and active view selection, with the generative formulation additionally capturing diverse interaction patterns that reflect affordance ambiguity. Current limitations include dependence on initial affordance estimates for guiding view selection and the object-centric assumption that requires segmentation in multi-object scenes. Addressing these through error-aware view selection and joint multi-object reconstruction would broaden applicability. As the method reconstructs functional regions from limited views, applying predicted affordances to robotic manipulation for interaction planning on completed geometry is a natural next step.

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